

DATA ANALYTICS INTERNSHIP PROJECT REPORT

Submitted To : **Decode labs**

Submitted By

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Table of Contents

1. Introduction
 2. Project 1 – Data Cleaning & Preparation
 3. Project 2 – Exploratory Data Analysis (EDA)
 4. Project 3 – SQL Data Analysis
 5. Business Insights
 6. Conclusion
-

Introduction

The objective of this internship project is to perform data cleaning, exploratory data analysis, and SQL-based business analysis using an e-commerce dataset. The project was completed using Microsoft Excel and Oracle SQL*Plus.

Tools Used:

- Microsoft Excel
- Oracle SQL*Plus

Dataset Used:

- E-commerce Order Dataset
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Project 1 – Data Cleaning & Preparation

Objective

The objective of Project 1 is to clean the raw dataset and prepare it for analysis.

Desc :-

The raw dataset contained missing values, duplicate records, inconsistent formatting, and invalid data entries. These issues can affect data accuracy and make it difficult to generate meaningful business insights. Data cleaning was performed to improve the quality, consistency, and reliability of the dataset before analysis. After cleaning the dataset, the data became more structured and suitable for exploratory data analysis and SQL-based reporting

Steps Performed :-

1. Imported Dataset

- Opened the raw dataset in Microsoft Excel.
- Converted the dataset into table format.

2. Checked Missing Values

- Identified blank cells.
- Replaced missing values appropriately.

3. Removed Duplicate Records

- Removed duplicate rows.
- Verified unique Order IDs.

4. Corrected Data Formats

- Standardized date formats.
- Corrected numeric formats.

5. Validated Dataset

- Verified TotalPrice values.
 - Removed invalid entries.
-

Project 1 Screenshots

OrderID	Date	Month	CustomerID	Product	Quantity	UnitPrice	TotalPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsinCart
ORD200000	1/4/2023	January	C72649	Monitor	5	570.62	2853.10	928 Main St	Debit Card	Shipped	TRK37947903	
ORD200001	8/23/2024	August	C75739	Phone	2	151.35	302.70	823 Main St	Online	Shipped	TRK91186779	
ORD200002	2/27/2024	February	C81728	Tablet	5	550.68	2753.40	512 Main St	Credit Card	Cancelled	TRK42903982	
ORD200003	10/15/2023	October	C33540	Chair	1	273.19	273.19	275 Main St	Debit Card	Returned	TRK62788070	
ORD200004	5/8/2025	May	C81840	Printer	4	626.01	2504.04	668 Main St	Online	Delivered	TRK29241424	
ORD200005	10/23/2023	October	C37249	Phone	2	245.86	491.72	934 Main St	Credit Card	Shipped	TRK72976927	
ORD200006	6/17/2025	June	C83492	Laptop	1	664.42	664.42	986 Main St	Gift Card	Returned	TRK96417362	
ORD200007	5/12/2023	May	C41460	Monitor	5	149.55	747.75	706 Main St	Cash	Shipped	TRK78809193	
ORD200008	4/2/2025	April	C26817	Phone	2	134.28	268.56	904 Main St	Gift Card	Cancelled	TRK1042692	
ORD200009	11/21/2023	November	C31946	Desk	4	509.38	2037.52	102 Main St	Credit Card	Shipped	TRK33478363	
ORD200010	12/29/2023	December	C43443	Tablet	5	625.97	3129.85	333 Main St	Credit Card	Returned	TRK98859248	
ORD200011	2/17/2024	February	C93861	Monitor	3	49.14	147.42	831 Main St	Online	Returned	TRK48234646	
ORD200012	10/15/2024	October	C38785	Monitor	2	180.50	361.00	179 Main St	Debit Card	Pending	TRK20419991	
ORD200013	8/30/2023	August	C88348	Laptop	3	201.49	604.47	903 Main St	Credit Card	Returned	TRK81913874	
ORD200014	3/27/2023	March	C98474	Tablet	2	393.33	786.66	980 Main St	Debit Card	Pending	TRK79186539	
ORD200015	7/17/2023	July	C39416	Printer	1	473.96	473.96	942 Main St	Cash	Delivered	TRK54930938	
ORD200016	1/24/2023	January	C28976	Printer	2	533.81	1067.62	914 Main St	Credit Card	Pending	TRK13194259	
ORD200017	3/2/2024	March	C60672	Tablet	1	423.40	423.40	382 Main St	Debit Card	Shipped	TRK11972116	
ORD200018	3/26/2025	March	C84329	Desk	1	431.44	431.44	891 Main St	Gift Card	Pending	TRK65191045	
ORD200019	1/23/2023	January	C14552	Monitor	5	224.00	1120.00	273 Main St	Cash	Returned	TRK59101365	
ORD200020	5/8/2023	May	C54829	Chair	5	62.51	312.55	587 Main St	Cash	Pending	TRK16718651	
ORD200021	11/17/2024	November	C99023	Monitor	4	342.95	1371.80	569 Main St	Cash	Delivered	TRK52646983	
ORD200022	1/29/2023	January	C34416	Monitor	2	49.62	99.24	633 Main St	Online	Pending	TRK75937350	
ORD200023	4/9/2025	April	C42372	Phone	4	273.25	1093.00	755 Main St	Debit Card	Cancelled	TRK36168083	
ORD200024	10/30/2024	October	C17470	Laptop	2	230.95	461.90	956 Main St	Online	Delivered	TRK28775057	
ORD200025	7/24/2024	July	C42577	Monitor	1	399.98	399.98	709 Main St	Debit Card	Delivered	TRK69423327	

Project 2 – Exploratory Data Analysis (EDA)

Objective

The objective of Project 2 is to analyze the cleaned dataset using pivot tables, charts, and statistical analysis.

Pivot Table Analysis

1. Product Revenue Analysis

- Created pivot table using Product and TotalPrice.
- Identified highest revenue-generating products.

2. Payment Method Analysis

- Analyzed most used payment methods.

3. Order Status Analysis

- Analyzed shipped, cancelled, and returned orders.

4. Monthly Sales Analysis

- Created monthly sales trend analysis.

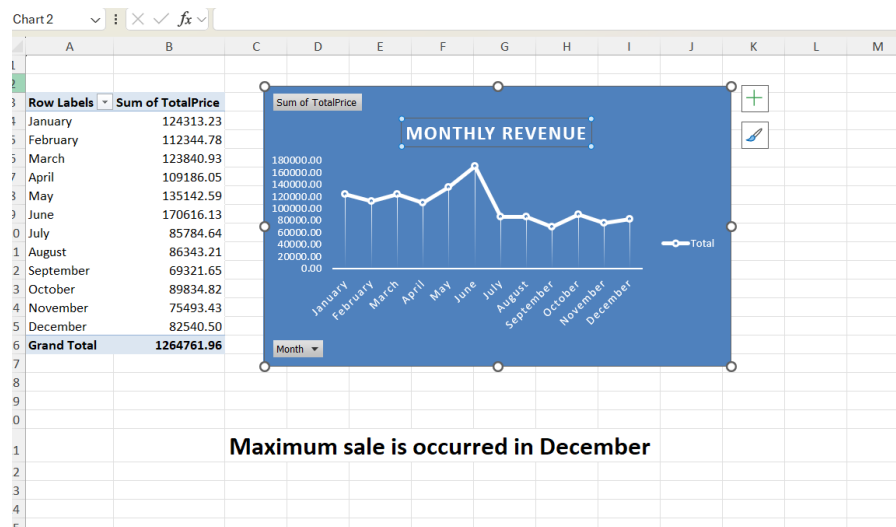
5. Referral Source Analysis

- Identified best marketing source.

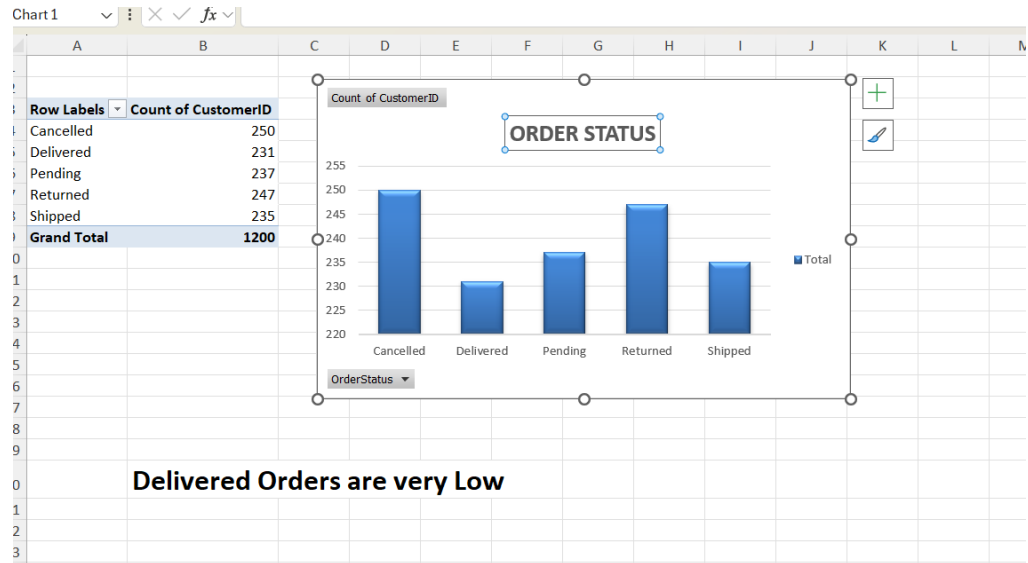
Charts Created

1. Product Revenue Bar Chart
 2. Monthly Sales Line Chart
 3. Payment Method Pie Chart
 4. Order Status Column Chart
 5. Referral Source Bar Chart
-

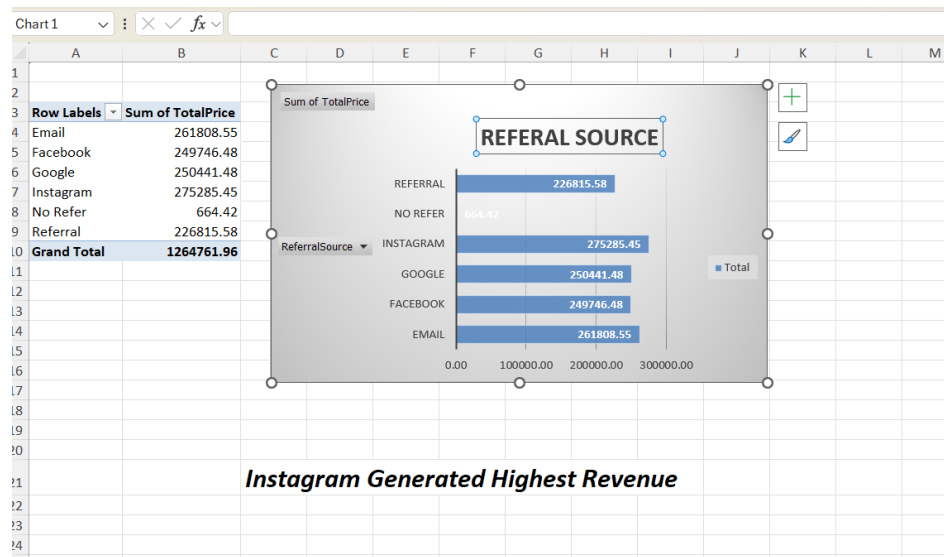
Project 2 Screenshots



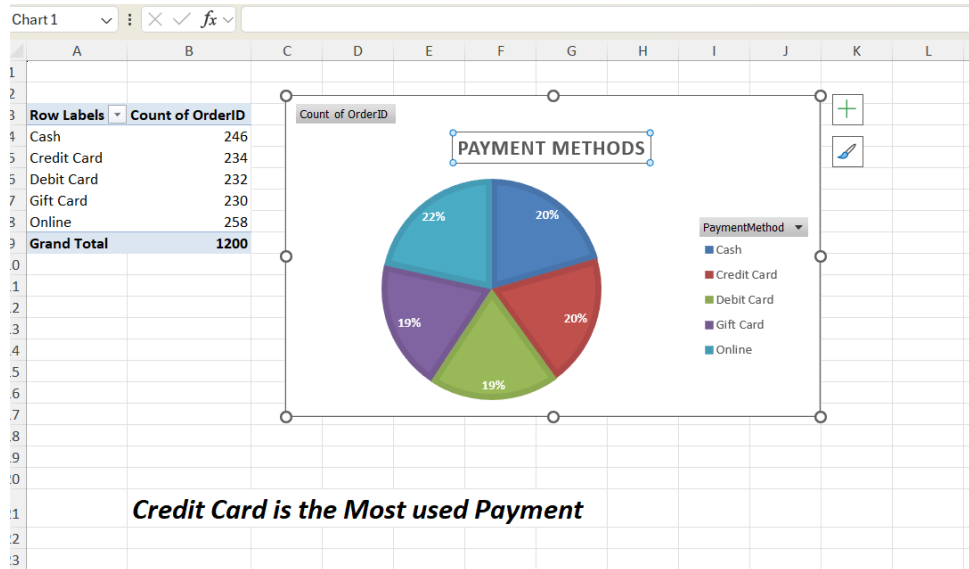
Revenue Chart



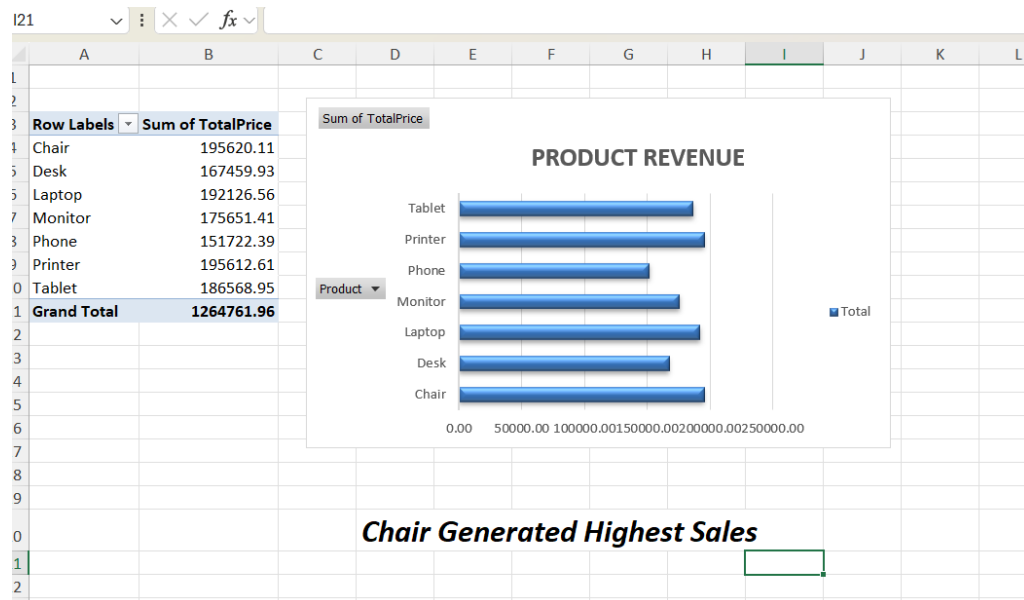
Delivery Order Status



Referral Source



Payment Method Status



Product Sales Status

Project 3 – SQL Data Analysis

Objective

The objective of Project 3 is to perform SQL-based business analysis using Oracle SQL*Plus.

SQL Queries Performed

1. SELECT Query

Used to retrieve records from the dataset.

2. WHERE Clause

Used to filter records.

3. ORDER BY

Used to sort records.

4. GROUP BY

Used for grouped analysis.

5. COUNT()

Used to count records.

6. SUM()

Used to calculate total revenue.

7. AVG()

Used to calculate average order value.

8. HAVING Clause

Used to filter grouped data.

SQL Output Screenshots

Oracle SQL*Plus

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ORD200961 11-MAY-24 C55464 Phone
ORD200962 06-SEP-24 C12459 Phone
ORD200963 11-MAR-24 C88075 Chair
ORD200964 20-AUG-23 C18583 Phone
ORD200965 18-JUN-25 C70412 Laptop
ORD200966 20-MAY-25 C58503 Monitor
ORD200967 06-MAY-25 C52879 Tablet
ORD200968 08-FEB-24 C67870 Monitor
ORD200969 05-JUN-25 C19430 Desk
ORD200970 07-SEP-24 C72664 Desk
ORD200971 27-MAY-24 C32650 Monitor

```

1200 rows selected.

```

SQL> SELECT *
2 FROM orders
3 WHERE OrderStatus = 'Shipped';

```

ORDERID	ORDERDATE	CUSTOMERID	PRODUCT	QUANTITY	UNITPRICE	TOTALPRICE	SHI
ORD200069	11-MAY-23	C43338	Monitor	3	478.65	1435.95	555
ORD200079	10-MAR-25	C76700	Tablet	2	311.91	623.82	105
ORD200081	08-JUN-23	C29314	Monitor	2	394.3	788.6	494
ORD200082	05-JAN-24	C65799	Tablet	4	565.2	2260.8	971
ORD200084	28-AUG-23	C64160	Monitor	2	165.49	330.98	967
ORD200087	10-OCT-24	C91245	Tablet	3	225.11	675.33	915
ORD200091	15-MAR-23	C98216	Laptop	5	492.75	2463.75	194
ORD200092	10-JUN-24	C29568	Printer	4	48.61	194.44	205
ORD200101	25-MAY-23	C15283	Printer	1	381.02	381.02	152
ORD200107	27-MAR-23	C16775	Printer	5	670.75	3353.75	444
ORD200109	17-SEP-23	C51157	Desk	3	592.87	1778.61	131
ORD200113	17-JAN-24	C65905	Laptop	5	190.08	950.4	405
ORD200115	27-MAY-23	C32580	Phone	5	374.99	1874.95	122
ORD200118	09-OCT-23	C27890	Printer	5	502.71	2513.55	800
ORD200119	11-JUN-25	C14683	Desk	2	534.67	1069.34	187
ORD200122	15-APR-24	C96787	Laptop	2	533.02	1066.04	588
ORD200125	06-MAY-24	C91899	Phone	2	512.65	1025.3	758
ORD200126	03-MAY-25	C55606	Desk	4	275.5	1102.35	
ORD200127	10-AUG-23	C71486	Desk	3	469.32	1407.96	365
ORD200129	27-OCT-23	C52751	Laptop	4	263.93	1055.72	427
ORD200132	28-MAR-23	C96257	Tablet	4	450.75	1803.58	
ORD200135	04-AUG-23	C79909	Printer	2	68.81	137.62	642
ORD200143	13-JUN-24	C66521	Chair	4	400.07	1600.28	935
ORD200155	27-FEB-23	C62740	Phone	1	424.4	424.4	444
ORD200181	31-AUG-24	C38966	Printer	5	404.39	2021.95	247
ORD200185	02-FEB-25	C26677	Laptop	1	218.7	218.7	572
ORD200188	15-MAR-25	C97715	Chair	2	363.91	727.82	927
ORD200194	08-JUN-24	C53501	Monitor	1	249.57	249.57	895
ORD200201	18-JUN-23	C10126	Monitor	2	674.9	1349.8	144
ORD200203	21-JAN-25	C99173	Printer	3	356.84	1070.52	175

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Search

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Oracle SQL*Plus

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```

SQL> SELECT TO_CHAR(OrderDate,'MONTH') AS Month,
2 SUM(TotalPrice) AS MonthlySales
3 FROM orders
4 GROUP BY TO_CHAR(OrderDate,'MONTH')
5 ORDER BY MonthlySales DESC;

```

MONTH	MONTHLYSALES
JUNE	170616.13
MAY	135142.59
JANUARY	124313.23
MARCH	123840.93
FEBRUARY	112344.78
APRIL	109186.05
OCTOBER	89834.82
AUGUST	86343.21
JULY	85784.64
DECEMBER	82540.5
NOVEMBER	75409.43
SEPTEMBER	69321.65

12 rows selected.

```

SQL> SELECT OrderStatus,
2 COUNT(*) AS Total
3 FROM orders
4 GROUP BY OrderStatus;

```

ORDERSTATUS	TOTAL
Delivered	231
Shipped	295
Cancelled	250
Returned	247
Pending	237

```

SQL> SELECT PaymentMethod,
2 COUNT(*) AS TotalOrders
3 FROM orders
4 GROUP BY PaymentMethod;

```

PAYMENTMETHOD	TOTALORDERS
Online	258
Credit Card	234
Cash	246
Gift Card	230
Debit Card	232

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```
Oracle SQL*Plus
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4 FROM orders
5 ORDER BY TotalPrice DESC
6 )
7 WHERE ROUNDH <= 5;

ORDERID      ORDERDATE  CUSTOMERID  PRODUCT                                QUANTITY  UNITPRICE  TOTALPRICE  SHI
-----
ORD200789    17-AUG-23  C57276      Tablet                                5         691.28     3456.4 185
ORD201122    07-JUN-23  C38840      Monitor                               5         678.19     3390.95 746
ORD200632    02-MAY-23  C67260      Laptop                               5         678.16     3390.8 465
ORD200469    26-NOV-23  C13877      Chair                                5         676.98     3384.9 895
ORD200328    28-FEB-23  C18404      Tablet                                5         674.04     3370.2 546

SQL> SELECT SUM(Quantity) AS TotalQuantitySold
2 FROM orders;

TOTALQUANTITYSOLD
-----
3535

SQL> SELECT AVG(Quantity) AS AvgQuantity
2 FROM orders;

AVGQUANTITY
-----
2.94583333

SQL> SELECT AVG(TotalPrice) AS AverageOrderValue
2 FROM orders;

AVERAGEORDERVALUE
-----
1053.9683

SQL> SELECT SUM(TotalPrice) AS TotalRevenue
2 FROM orders;

TOTALREVENUE
-----
1264761.96

SQL> SELECT COUNT(*) AS CancelledOrders
2 FROM orders
3 WHERE OrderStatus = 'Cancelled';

CANCELLEDORDERS
-----
250

SQL> |
```

Business Insights

1. Instagram generated the highest revenue.
2. Chair generated the highest product revenue.
3. Credit Card was the most used payment method.
4. Maximum order value was identified successfully.
5. Monthly sales trends were analyzed using pivot tables.

Conclusion

The internship project was successfully completed using Excel and Oracle SQL*Plus. The project included data cleaning, exploratory data analysis, and SQL-based business analysis. Meaningful business insights were generated from the dataset.

References

- Microsoft Excel
- Oracle SQL*Plus
- Decodelabs Internship Dataset